

DAKAR, Senegal

WMO: 616410

Lat: 14.740N

Lon: 17.490W

Elev: 26

StdP: 101.01

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.9	17.2	1.0	4.1	25.5	5.0	5.4	25.0	10.3	21.1	9.5	21.5	4.7	0	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	5.6	32.8	22.2	31.6	24.7	31.0	25.4	28.0	29.9	27.5	29.5	27.1	29.3	3.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	29.1	27.0	22.8	28.8	26.5	22.1	28.6	90.2	29.7	87.5	29.8	85.6	29.3	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.4	7.7	15.6	36.8	1.2	1.4	14.7	37.9	14.0	38.7	13.3	39.6	12.5	40.6		
(5)			9.7	29.1	1.5	0.9	8.6	29.7	7.7	30.2	6.8	30.7	5.7	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	22.4	21.7	21.9	22.1	23.5	26.1	27.5	27.7	27.9	28.2	26.8	24.6		
	DBStd	2.91	2.10	2.15	1.87	1.45	1.42	1.28	0.94	0.94	1.01	1.11	1.44	1.80		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5497	383	329	369	364	418	483	543	549	537	565	505	452		
	CDD18.3	2456	125	96	112	114	159	233	285	291	287	307	255	193		
	CDH23.3	19692	413	316	336	257	571	1787	2941	3177	3179	3466	2268	982		
	CDH26.7	5303	92	85	76	32	51	323	741	889	988	1196	624	207		
(14) Wind	WSAvg	4.2	4.9	5.2	5.4	5.5	4.8	3.6	3.4	3.2	2.8	3.2	4.2	4.7		
(15) Precipitation	PrecAvg	425	0	1	0	0	0	8	59	169	137	31	1	1		
	PrecMax	903	12	40	9	0	1	76	220	413	370	183	20	19		
	PrecMin	116	0	0	0	0	0	0	0	20	40	0	0	0		
	PrecStd	172	2	6	1	0	0	14	51	92	73	43	3	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.1	32.9	30.0	29.2	31.0	31.3	31.2	32.1	33.4	34.8	33.0		
		MCWB	16.5	18.1	17.1	20.5	23.3	25.2	25.8	26.3	26.8	25.2	20.3	18.2		
	2%	DB	29.1	29.0	28.2	27.0	28.0	30.0	30.8	30.8	31.2	32.1	32.1	30.2		
		MCWB	17.2	17.5	18.7	21.1	22.9	25.0	25.7	26.4	26.9	26.2	22.3	19.5		
	5%	DB	27.0	26.8	26.3	25.7	27.0	29.1	30.1	30.2	30.8	31.2	30.9	28.9		
		MCWB	17.8	17.8	19.3	21.0	22.7	24.7	25.5	26.2	26.9	26.4	23.2	19.8		
	10%	DB	25.2	25.0	24.9	24.8	26.0	28.4	29.6	29.9	30.1	30.9	29.8	27.3		
		MCWB	18.0	17.9	19.3	20.9	22.3	24.4	25.3	26.1	26.7	26.4	23.3	20.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	22.0	22.5	23.2	25.2	26.7	27.2	28.3	28.8	28.7	26.8	25.0		
		MCDB	24.9	25.2	25.4	25.8	27.5	28.8	29.3	29.7	30.4	30.8	29.2	27.0		
	2%	WB	21.2	20.9	21.6	22.2	24.1	26.1	26.7	27.5	28.0	28.0	26.1	23.9		
		MCDB	23.8	23.9	24.4	24.9	26.5	28.3	29.0	29.2	29.8	30.2	28.7	26.0		
	5%	WB	20.6	20.3	20.9	21.7	23.3	25.5	26.2	27.0	27.5	27.5	25.6	23.2		
		MCDB	23.3	23.0	23.7	24.3	25.7	27.9	28.7	29.0	29.5	29.7	28.2	25.5		
	10%	WB	19.9	19.7	20.4	21.1	22.7	25.0	25.9	26.6	27.1	27.0	25.0	22.5		
		MCDB	22.8	22.3	23.1	23.7	25.1	27.6	28.4	28.7	29.2	29.4	27.6	25.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	6.8	6.4	5.6	5.0	4.7	4.5	4.5	5.1	5.6	6.2	6.3		
		MCDBR	9.4	10.5	9.2	7.0	5.9	5.2	5.0	4.8	5.6	6.2	8.8	8.9		
	5% WB	MCWBR	5.9	4.4	3.6	3.1	2.8	2.6	2.7	2.8	3.2	3.3	5.0	6.1		
		MCWBR	7.0	6.7	6.5	6.0	5.3	4.9	4.7	4.7	5.2	5.7	6.6	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.529	0.551	0.616	0.630	0.687	0.739	0.692	0.621	0.595	0.601	0.523	0.509		
		taud	1.911	1.869	1.743	1.769	1.661	1.640	1.746	1.912	1.957	1.907	2.025	2.012		
	Ebn at Noon		764	769	732	720	670	628	661	717	738	720	765	770		
		Edh at Noon	188	204	237	231	253	255	230	197	189	194	166	166		
(44) All-Sky Solar Radiation	RadAvg		5.23	5.83	6.79	7.15	7.07	6.60	5.91	5.67	5.56	5.75	5.22	4.92		
	RadStd		0.34	0.36	0.24	0.21	0.26	0.15	0.24	0.27	0.24	0.29	0.31	0.23		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page